Orion Robotics

Orion A1 – Motion Controller Firmware Guide

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The Orion A1 motion controller firmware orchestrates 22 torque-controlled actuators through high-frequency sensor feedback. Each joint receives millisecond-level torque corrections based on encoder and IMU inputs, stabilising posture and gait transitions even under uneven loads. The controller continuously adjusts PID coefficients in real time to maintain balance while adhering to torque envelopes.

As illustrated on the right, the firmware follows a closed-loop flow: sensors feed a Kalman-based state estimator, which refines motion data before forwarding it to the PID controller. Commands are transmitted to motor drivers, while encoders provide feedback for precise error correction. This structure allows Orion A1 to walk fluidly, react to collisions, and recover from instability with minimal delay.

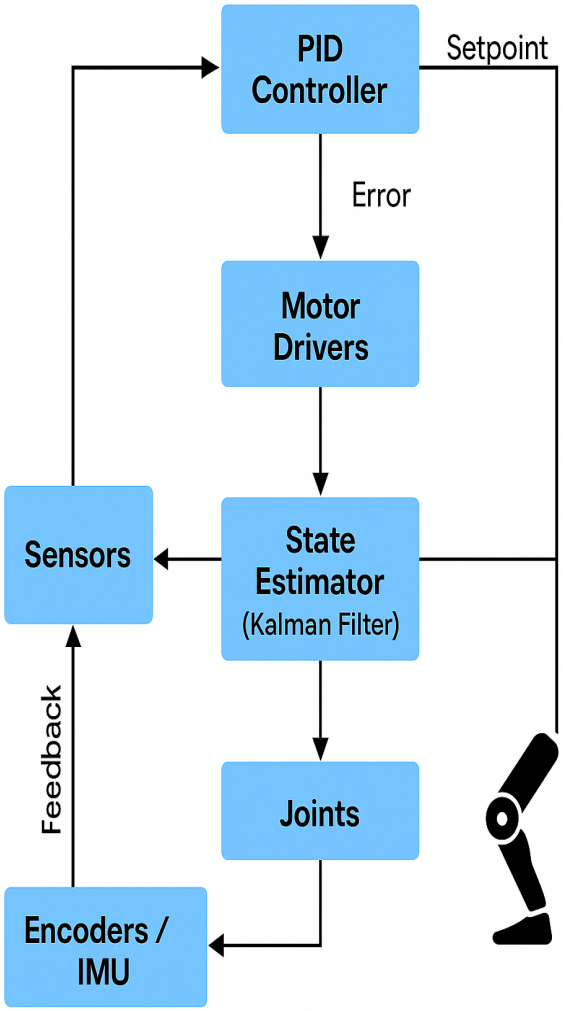

Figure 1. Orion A1 motion control loop schematic.

1. Firmware Architecture and Tuning

Threads include Motion Planning, Sensor Fusion, Actuation, and Safety Monitoring. The Safety thread runs on an isolated co-processor, able to cut power in under 200 ms when over-torque or drift anomalies are detected. A deterministic message bus coordinates inter-thread communication to eliminate timing jitter.

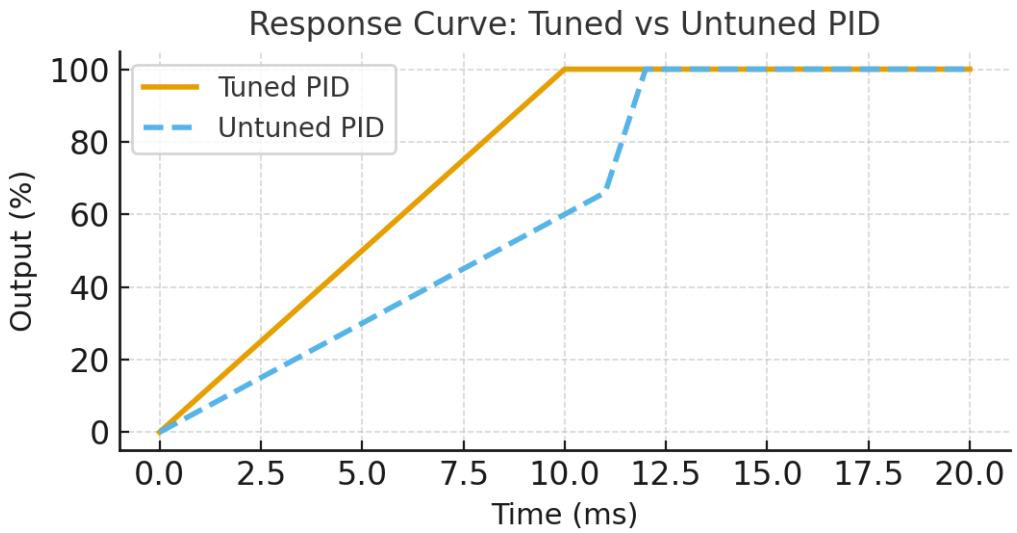

Figure 2. PID tuning comparison between tuned and untuned systems.

Tuning workflow: start with Kp to improve response, add Kd to reduce overshoot, and finally apply a minimal Ki to eliminate steady-state error. Recalibrate IMU offsets before any tuning session.

2. Diagnostics and Performance

The diagnostics module logs telemetry at 100 Hz intervals, including torque variance, IMU drift, CPU utilisation, and temperature metrics. Logs can be exported from the Orion Engineering Console for analysis.

Timestamp (ms)	Parameter	Unit	Nominal Range	Warning	Notes
0–1000	Torque Feedback	Nm	0–35	>40	Investigate joint friction.
0–1000	IMU Drift	°/s	0–0.8	>0.8	Recalibrate IMU offsets.
0–1000	Control Loop Jitter	ms	0–1.0	>1.5	Reduce non-critical tasks.
0–1000	CPU Load	%	30–60	>80	Optimise PID computations.
0–1000	Temperature	°C	25–55	>60	Activate cooling system.

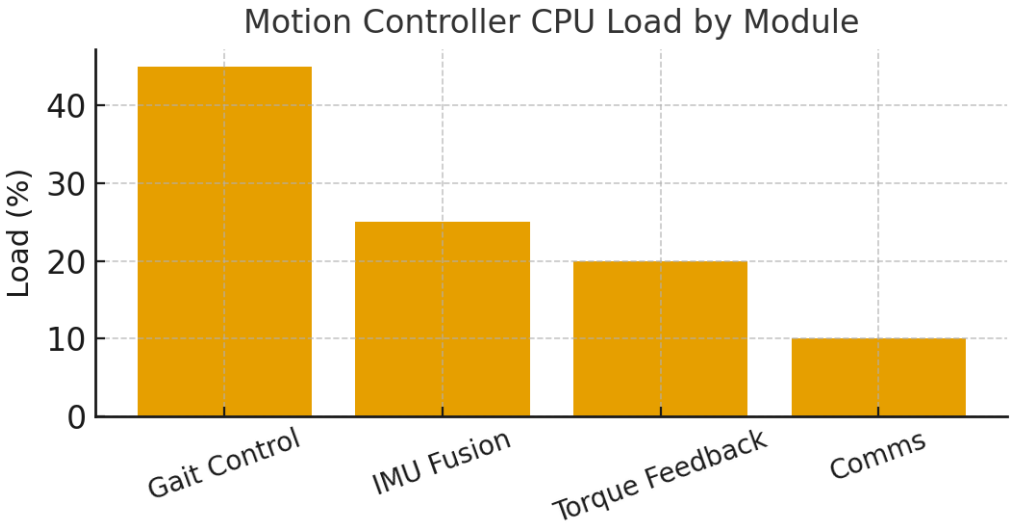

Figure 3. CPU load distribution by firmware module.

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